

On the Structural Origin of Cosmological Parameter Tensions and Transport-Induced Divergence

An AASC Closure Theorem for Cosmological Inference

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Abstract

Cosmological parameter tensions, most notably the discrepancy in the Hubble constant inferred from early- and late-universe observations, are typically treated as empirical anomalies requiring new dynamics, modified gravity, or additional degrees of freedom. This paper proves that such tensions arise as a structurally necessary consequence of transport-induced divergence in cosmological inference when admissibility-preserving conditions are not fully satisfied.

We formalize cosmological inference as an external data–model–parameter–projection tuple and define an admissibility-preserving transport map linking early- and late-epoch inference regimes. A Cosmological Degrees-of-Freedom Lemma shows that any admissibility-relevant modification of this tuple must occur in a finite set of coordinates. From this, a Cosmological Divergence Theorem is derived: any failure of admissibility-preserving transport produces a divergence in inferred parameters, including the expansion rate.

The proof proceeds by constraint-induced exhaustion. All divergence modes are mapped into a finite normal-form family and shown to collapse under admissibility constraints, yielding a closure result: no same-domain inference scheme can eliminate such tensions without either changing scope, importing additional structure, or restoring admissibility-preserving transport. A concrete two-epoch toy model demonstrates explicit parameter mismatch under tensor-witness omission and its resolution under a completed admissibility-preserving realization.

The result reframes cosmological parameter tensions as transport-induced structural effects rather than failures of underlying physics. It establishes that standard Lambda-CDM implicitly combines early- and late-time projections without enforcing a single admissibility-preserving transport, and therefore cannot eliminate such tensions within its current formulation. The appropriate replacement is a constraint-native cosmological realization in which gravitational and quantum sectors are treated as bookkeeping projections of a common admissibility-bearing quotient architecture, ensuring consistency across all epochs by construction.

Scope note. The phrase “Hubble tension” is used in its standard observational sense: early-universe inference under a cosmological model gives a lower value of H_0 than many late-universe distance-ladder measurements. The numerical values often quoted in the literature, roughly 67 and $73 \text{ km s}^{-1} \text{ Mpc}^{-1}$, are orientation data rather than premises of the theorem. The result below concerns typed comparison of early and late expansion inferences under a fixed admissibility-bearing cosmological scope.

1 Problem and claim

The Hubble tension is usually formulated as a conflict between two inference pipelines. CMB-based inference, using a cosmological model, yields one estimate of the present expansion rate;

local distance-ladder inference yields another. Standard responses therefore search for a missing dynamical ingredient, modified gravity, altered early-universe physics, calibration error, estimator bias, or some combination of these. All such options remain open as empirical model-building strategies. The present paper does not compete with them at that level. It asks what must be true before two such inferences are even admissible as estimates of one and the same cosmological object.

That question is structural. An expansion estimate is not merely a number attached to a symbol H_0 . It is the output of a typed inference: data are selected, records are calibrated, a model family is fixed, parameters are extracted, nuisance structure is marginalized or transported, and an error discipline declares what counts as agreement. If an early-universe estimate and a late-universe estimate are asserted to concern the same expansion target, then these ledgers must commute. If they do not commute, the disagreement is not a harmless change of presentation; it marks a typed divergence in the cosmological realization certificate.

The paper proves this claim as an AASC closure theorem. It first defines the external cosmological inference tuple in ordinary mathematical-physics terms, without using AASC vocabulary. It then proves that every Hubble-relevant first divergence must occur in one of finitely many external degrees of freedom: data/records, model content, parameter target, prediction map, inference map, transport, or error discipline. Only after this external exhaustion is established are the coordinates translated into the AASC ledger: anchors, tensor witnesses, quotient identity, continuation, boundary records, estimator structure, and error envelope.

Main theorem, informally. If a cosmological inference tuple is equipped with a complete same-domain AASC realization certificate, then early and late expansion inferences commute wherever the certified transport is defined. Therefore, if a persistent Hubble-tension witness remains while both estimates are claimed to concern the same fixed expansion target, the transport has a nonempty first-divergence image in a finite family of typed loci. The mismatch must be anchor-level, tensor-level, quotient-level, continuation-level, boundary-record-level, estimator-level, or error-envelope-level. It cannot be merely skin.

Why this is not a numerical cosmology claim. The theorem does not predict the size of the observed tension and does not identify the physical mechanism responsible for it. It fixes the admissibility burden on any proposed resolution. A successful resolution must not merely shift a value of H_0 ; it must supply a revised certificate showing how early and late expansion claims are transported as the same standing-bearing target under the declared error discipline.

Relation to AASC. The AASC kernel supplies admissibility, standing, reference, and irreversibility as the conditions under which determinate same-domain construction is evaluable. The AASC interface supplies AMetricity, bivalence, unique admissible interior, and standing conservation. The GR/QM projection architecture supplies the background reading of GR and QM as bookkeeping projections of a common admissibility-bearing quotient architecture. The present paper applies that machinery to cosmological inference: it shows that the Hubble tension, if persistent under a same-target claim, is a transport-divergence theorem object rather than a mere disagreement of names.

2 AASC background and projection architecture

This section states only the dependency structure used below. The paper does not re-prove the AASC kernel, the admissibility-interface theorem, or the GR/QM projection theorem.

Definition 2.1 (AASC kernel). *The AASC kernel is the four-role package*

$$K = (\text{admissibility}, \text{standing}, \text{reference}, \text{irreversibility}).$$

A domain instantiates the kernel when it supports identity-preserving, admissibility-assessable construction of determinate same-domain objects under non-repairable commitment.

Definition 2.2 (AASC interface). *The AASC interface is the fixed-domain consequence package consisting of:*

1. *AMetric pre-admissible boundary;*
2. *bivalent evaluated admissibility;*
3. *standing as reuse-stable admissibility;*
4. *unique admissible interior up to identity-preserving equivalence;*
5. *conservation of standing under same-scope continuation.*

Definition 2.3 (Admissibility-bearing quotient architecture). *An admissibility-bearing quotient architecture is a structure*

$$\mathcal{A} = (\text{Anc}, \text{Ten}, Q, \text{Cont}, \text{Bnd}, \text{Est}, \text{Err})$$

where Anc are domain-fixing anchors, Ten are invariant tensor witnesses, Q is the standing quotient, Cont is admissible continuation, Bnd is boundary/record structure, Est is the admissible estimator family, and Err is the declared error discipline. A realization $\mathcal{R}$ of $\mathcal{A}$ supplies presentation data whose standing-bearing content factors through this tuple.

Definition 2.4 (GR and QM projections). *A GR projection $\Pi_{\text{GR}}(\mathcal{R})$ is the constrained-Hamiltonian interior realization of $\mathcal{R}$: standing surface, physical quotient, characteristic redescription, first-class closure, and multiplier-form Hamiltonian dynamics. A QM projection $\Pi_{\text{QM}}(\mathcal{R})$ is the Hilbert/Schroedinger interior together with Bell-scope and measurement-record boundary data. Neither projection is primitive relative to $\mathcal{A}$; each is a role-specific bookkeeping projection of the same constraint architecture.*

Remark 2.5 (Why projection matters). *Ordinary cosmological bookkeeping often treats the GR projection and the quantum matter projection as independent primitives coupled by an equation. Under AASC, that posture is not fundamental. Coupling projections does not replace construction of the common quotient architecture from which both projections are realized. This is the structural setting in which cosmological transport divergence becomes meaningful.*

3 External cosmological inference object

We now define cosmological comparison before using AASC categories. This makes the divergence theorem a theorem about the ordinary inference problem rather than a post hoc reclassification.

Definition 3.1 (Cosmological inference tuple). *A cosmological inference tuple is*

$$\mathcal{I} = (\mathcal{D}_E, \mathcal{D}_L, \mathcal{M}, \Theta_E, \Theta_L, \mathcal{P}_E, \mathcal{P}_L, I_E, I_L, \mathcal{R}_E, \mathcal{R}_L, \text{Err}_E, \text{Err}_L),$$

where:

1. $\mathcal{D}_E$ and $\mathcal{D}_L$ are early- and late-universe data spaces;
2. $\mathcal{M}$ is a model family, such as FLRW plus specified matter, radiation, dark-energy, perturbation, and calibration structure;
3. Θ_E and Θ_L are parameter spaces containing an expansion coordinate called H_0 ;
4. $P_E : \mathcal{M} \times \Theta_E \rightarrow \mathcal{D}_E$ and $P_L : \mathcal{M} \times \Theta_L \rightarrow \mathcal{D}_L$ are projection/prediction maps;
5. $I_E : \mathcal{D}_E \rightarrow \Theta_E$ and $I_L : \mathcal{D}_L \rightarrow \Theta_L$ are inference maps, possibly set-valued or distribution-valued;
6. R_E and R_L are record structures: calibration records, selection functions, likelihood-construction records, and boundary conditions used by the two pipelines;
7. Err_E and Err_L are statistical and systematic error disciplines.

The tuple is external: it is stated in ordinary mathematical-physics terms before being translated into the AASC ledger.

Definition 3.2 (Expansion extraction). *Let*

$$\eta_E : \Theta_E \rightarrow \mathbb{R}, \quad \eta_L : \Theta_L \rightarrow \mathbb{R}$$

extract the expansion coordinate H_0 . If I_E, I_L are distribution-valued, $\eta_E \circ I_E$ and $\eta_L \circ I_L$ denote the induced reported estimate, interval, posterior, or confidence/credible object according to the declared error discipline.

Definition 3.3 (Expansion target). *An expansion target $\mathbf{H}$ is the same-domain object represented by the present-expansion parameter after quotienting skin-level parameterization. Two estimates $h_E \in \Theta_E$ and $h_L \in \Theta_L$ are estimates of the same target only if there is an admissibility-preserving transport identifying their quotient roles.*

Definition 3.4 (External transport). *An external early-to-late transport for $\mathcal{I}$ is a tuple*

$$\mathcal{T}_{E \rightarrow L} = (T_D, T_\Theta, T_M, T_R, T_{\text{Err}}),$$

where $T_D : \mathcal{D}_E \dashrightarrow \mathcal{D}_L$ is a data/witness transport, $T_\Theta : \Theta_E \dashrightarrow \Theta_L$ is parameter transport, T_M is model-content transport, $T_R : R_E \dashrightarrow R_L$ is record transport, and $T_{\text{Err}} : \text{Err}_E \dashrightarrow \text{Err}_L$ is error-envelope transport. The transport claims that early-universe standing-bearing content is carried into late-universe standing-bearing content as the same cosmological target.

Definition 3.5 (Commuting expansion inference). *Early and late expansion inference commute on $d_E \in \mathcal{D}_E$ if*

$$\eta_L(I_L(T_D(d_E))) \sim_{\text{Err}} \eta_L(T_\Theta(I_E(d_E))),$$

where $\sim_{\text{Err}}$ means equality of the expansion target within the transported joint error discipline. If the expressions are not of the same quotient type, commutation is undefined rather than false.

Definition 3.6 (Λ CDM as an external inference object). *For this paper, Λ CDM denotes a restricted instance*

$$\mathcal{I}_{\Lambda\text{CDM}} = (\mathcal{D}_E, \mathcal{D}_L, \mathcal{M}_{\Lambda\text{CDM}}, \Theta_{\Lambda\text{CDM}}, \Theta_{\Lambda\text{CDM}}, P_E^{\Lambda\text{CDM}}, P_L^{\Lambda\text{CDM}}, I_E^{\Lambda\text{CDM}}, I_L^{\Lambda\text{CDM}}, R_E, R_L, \text{Err}_E, \text{Err}_L),$$

where $\mathcal{M}_{\Lambda\text{CDM}}$ contains the usual low-parameter FLRW background, cold dark matter, baryons, radiation, and a cosmological constant, together with the record and estimator structures used in the specified early and late pipelines. No theorem below assumes Λ CDM is false. It treats Λ CDM as a bookkeeping realization candidate and asks whether it supplies a complete same-target transport certificate.

External degrees of freedom. The following lemma is the external counterpart of the AASC normal-form map. It shows that the divergence coordinates used below are forced by the ordinary inference tuple itself.

Definition 3.7 (External degrees of freedom of cosmological inference). *For a cosmological inference tuple $\mathcal{I}$, write*

$$\text{DOF}(\mathcal{I}) = (\mathcal{D}_E, \mathcal{D}_L, \mathcal{M}, \Theta_E, \Theta_L, \eta_E, \eta_L, \mathbf{P}_E, \mathbf{P}_L, I_E, I_L, \mathbf{R}_E, \mathbf{R}_L, \text{Err}_E, \text{Err}_L).$$

A change of $\text{DOF}(\mathcal{I})$ is Hubble-relevant when it changes either the reported early expansion object $\eta_E(I_E(d_E))$, the reported late expansion object $\eta_L(I_L(d_L))$, or the error-equivalence relation used to compare them.

Lemma 3.8 (Cosmological degrees-of-freedom lemma). *Every Hubble-relevant change of a cosmological inference tuple changes at least one of the following external loci:*

- G_1 : data/record spaces $(\mathcal{D}_E, \mathcal{D}_L, \mathbf{R}_E, \mathbf{R}_L)$,
- G_2 : model family $\mathcal{M}$,
- G_3 : parameter spaces and target extractors $(\Theta_E, \Theta_L, \eta_E, \eta_L)$,
- G_4 : prediction maps $(\mathbf{P}_E, \mathbf{P}_L)$,
- G_5 : inference maps (I_E, I_L) ,
- G_6 : external transport $\mathcal{T}_{E \rightarrow L}$,
- G_7 : error disciplines $(\text{Err}_E, \text{Err}_L)$.

If all seven loci are fixed up to presentation equivalence, then the early and late expansion reports and their equality-in-error verdict are fixed. Hence any remaining change is skin for the Hubble comparison.

Proof. The reported early object is obtained only from $d_E \in \mathcal{D}_E$, the record and prediction structures used to model it, the model family $\mathcal{M}$, the inference map I_E , and the extractor η_E . The late object is obtained analogously from $d_L \in \mathcal{D}_L$, $\mathcal{M}$, I_L , and η_L . Comparison of the two objects is determined by the external transport and the error disciplines. Therefore any change that leaves $G_1, \dots, G_7$ fixed leaves the reports and their comparison verdict fixed. A Hubble-relevant change must therefore occur in at least one G_i . Conversely, each G_i can change a report or the comparison verdict in ordinary inference practice, so the list is not a redundant enumeration of AASC categories but the coordinate decomposition of the external tuple itself. $\square$

Remark 3.9 (Structural rather than numerical theorem). *The closure theorem below identifies the necessary locus of a persistent discrepancy under a claimed same-target comparison. It does not derive the observed value, significance, or preferred mechanism of the Hubble tension. Numerical resolution requires a concrete cosmological realization certificate and empirical fit.*

4 AASC cosmological type discipline

The phrase “same target” is now made explicit as a type condition. This is the analogue of the same-domain type discipline in the quantum-gravity closure paper.

Definition 4.1 (Cosmological construction type). *A cosmological construction type is*

$$\tau_C = (\text{Anc}_C, \text{Ten}_C, Q_C, \text{Cont}_C, \text{Bnd}_C, \text{Est}_C, \text{Err}_C),$$

AASC role	Cosmological meaning	External loci	Divergence co-ordinates
Anchor	comparison domain, metric/FLRW regime, units, calibration anchors, epoch roles, declared scope	G_1, G_2	L_1
Tensor witness	standing expansion-relevant content: density sectors, curvature, perturbations, sound horizon, vacuum/dark-sector and quantum-matter witnesses	G_1, G_2, G_4	L_2
Quotient identity	same physical expansion target, not merely a shared parameter name	G_3, G_6	L_3
Continuation	early-to-late history, evolution, matching, and no post-hoc repair	G_2, G_6	L_4
Boundary record	calibration records, selection functions, covariance records, likelihood-construction records, boundary conditions	G_1	L_5
Estimator	priors, likelihoods, nuisance projections, estimator choices, reported-summary rules	G_5	L_6
Error envelope	statistical and systematic equality discipline	G_7	L_7

Table 1: Mapping from external cosmological inference degrees of freedom to AASC roles and divergence coordinates. This table grounds the AASC ledger in the ordinary inference tuple.

where anchors fix the domain, tensor witnesses fix standing-bearing cosmological content, Q_C fixes quotient identity, Cont_C fixes lawful early-to-late continuation, Bnd_C fixes boundary/record conditions, Est_C fixes admissible estimators, and Err_C fixes the error discipline.

Definition 4.2 (Same-domain well-typed transport). *An external transport $\mathcal{T}_{E \rightarrow L}$ is same-domain well-typed for τ_C iff it preserves every component of τ_C up to lawful equivalence. Thus it must preserve anchors, tensor witnesses, quotient identity, continuation, boundary records, estimator type, and error discipline. A transport that fails any component is not a same-domain map; it is a typed certificate revision or theorem-object change.*

Definition 4.3 (Cosmological realization certificate). *A cosmological realization certificate for $\mathcal{I}$ is*

$$\text{Cert}_C = (\tau_C, T_C)$$

where τ_C is a cosmological construction type and $T_C \neq \emptyset$ is the class of same-domain well-typed transports. A certificate is complete for an early/late comparison when it contains the transport needed to compare the two expansion inferences and proves commutation of that comparison.

Definition 4.4 (Hubble-tension witness). *A Hubble-tension witness is a pair $(d_E, d_L) \in \mathcal{D}_E \times \mathcal{D}_L$ such that both are claimed to estimate the same expansion target H , yet*

$$\eta_E(I_E(d_E)) \not\sim_{\text{Err}} \eta_L(I_L(d_L)).$$

The witness is persistent when the mismatch remains after the declared error discipline of $\mathcal{I}$ is applied.

5 Cosmological divergence theorem

This section proves the formal divergence result from the external inference tuple and the type discipline.

Definition 5.1 (Technical divergence coordinates). *Given $\mathcal{I}$ and Cert_C , a first same-domain divergence can occur only in one of the following coordinates:*

L_1 Anchor divergence:

domain anchors, calibration anchors, epoch labels, or comparison scope differ.

L_2 Tensor-witness divergence:

energy-density, curvature, equation-of-state, perturbation, vacuum/dark-energy, or quantum-sector witnesses differ or are incomplete.

L_3 Quotient-identity divergence:

early and late estimates are not estimates of the same quotient target.

L_4 Continuation divergence:

early-to-late evolution/transport differs, fails to close, or changes scope.

L_5 Boundary-record divergence:

observational records, calibration records, selection functions, or boundary conditions do not transport as the same standing content.

L_6 Estimator divergence:

inference maps, priors, parametrizations, likelihood choices, or nuisance projections perform standing-bearing work not fixed by the quotient ledger.

L_7 Error-envelope divergence:

statistical/systematic error discipline differs or is moved during comparison.

Lemma 5.2 (Coordinate exhaustion for cosmological inference). *Let $\mathcal{T}_{E \rightarrow L}$ be an external early-to-late transport for $\mathcal{I}$. If $\mathcal{T}_{E \rightarrow L}$ is not same-domain well-typed, then at least one coordinate $L_1, \dots, L_7$ is nontrivial. Conversely, if all coordinates are trivial, $\mathcal{T}_{E \rightarrow L}$ is same-domain well-typed.*

Proof. By the cosmological degrees-of-freedom lemma, every Hubble-relevant first divergence occurs in one of the external loci $G_1, \dots, G_7$. Table 1 identifies the AASC role carried by each such locus: anchors, tensor witnesses, quotient identity, continuation, boundary records, estimator structure, or error discipline. These are exactly the components of a same-domain cosmological construction type. Therefore failure of same-domain well-typedness has a nonempty image in $L_1, \dots, L_7$. Conversely, if every L_i is trivial, each component of the construction type is preserved; the transport is same-domain well-typed, and any remaining variation is skin for the comparison. $\square$

Definition 5.3 (First divergence index). *Fix the ordered coordinate list $(L_1, \dots, L_7)$. A first divergence index for a transport is the least i such that L_i is nontrivial, after quotienting away skin-level presentation changes. If no such i exists, the transport is skin-equivalent to a same-domain well-typed transport.*

Theorem 5.4 (Cosmological divergence theorem). *Let $\mathcal{I}$ be a cosmological inference tuple equipped with a complete realization certificate Cert_C . If $\mathcal{T}_{E \rightarrow L} \in T_C$, then early and late expansion inference commute wherever the transport is defined:*

$$\eta_L(I_L(T_D(d_E))) \sim_{\text{Err}} \eta_L(T_\Theta(I_E(d_E))).$$

Consequently, if a persistent Hubble-tension witness exists while both estimates are claimed to concern the same fixed expansion target, then the transport has a nonempty first-divergence image in $L_1, \dots, L_7$.

Proof. Because $\mathcal{T}_{E \rightarrow L} \in T_C$, it is same-domain well-typed. Therefore it preserves the anchors specifying the comparison domain, the tensor witnesses supplying expansion-relevant content, the quotient identity of the expansion target, the lawful continuation from early to late epochs, the boundary/record structures entering observation, the estimator type, and the error envelope. The late inference applied after transport and the transported early inference are thus two presentations of the same standing-bearing target under the same certificate. By quotient identity their expansion content agrees; by Err_C , this agreement is reported as $\sim_{\text{Err}}$. This proves commutation.

For the contrapositive, assume persistent noncommutation while same-target status is asserted. If all coordinates $L_1, \dots, L_7$ were trivial, the coordinate-exhaustion lemma would make the transport same-domain well-typed, and commutation would follow. Hence at least one coordinate is nontrivial. $\square$

Corollary 5.5 (No skin-only resolution). *A persistent Hubble-tension witness cannot be resolved by changing only skin-level presentation. A skin change preserves the construction type and therefore cannot change the same-target expansion inference outside the declared error envelope.*

Corollary 5.6 (No geometry-only same-scope repair). *If the GR projection is treated as an AER-rigid constrained-Hamiltonian interior, then a purely geometric same-scope adjustment that changes the inferred expansion history without changing tensor witnesses, quotient identity, continuation, or boundary data is not an admissible repair. It is either skin, which cannot resolve the witness, or theorem-object change.*

6 Normal forms and closure

The technical coordinates now translate into AASC normal forms. Because the coordinates are derived from the external inference tuple, this is not a post hoc taxonomy.

Definition 6.1 (Cosmological AASC normal forms). *A cosmological divergence has normal-form image in the following family:*

N_1 **Anchor motion:**

the domain or calibration anchors move.

N_2 **Tensor-witness change:**

the expansion-relevant witness ledger changes.

N_3 Quotient motion:

the identity of the expansion target changes.

N_4 Continuation failure:

early-to-late transport fails to preserve lawful continuation.

N_5 Boundary-record failure:

measurement or calibration records do not preserve standing.

N_6 Estimator selector:

an inference map, prior, parametrization, likelihood, or nuisance projection performs unlicensed standing-bearing work.

N_7 Error-envelope motion:

the error discipline changes during comparison.

Coordinate	Technical locus	Normal form	AASC reading
L_1	Domain, calibration, epoch, or scope anchor	N_1	anchor motion
L_2	Energy, curvature, equation of state, perturbation, vacuum/dark-energy, quantum witness	N_2	tensor-witness change
L_3	Same-target identity of H_0	N_3	quotient motion
L_4	Early-to-late evolution / transport	N_4	continuation failure
L_5	Records, calibrations, selection functions, boundary conditions	N_5	boundary-record failure
L_6	Inference maps, priors, parametrizations, likelihoods, nuisance projections	N_6	estimator selector
L_7	Statistical/systematic error model	N_7	error-envelope motion

Theorem 6.2 (Normal-form image theorem). *Every persistent Hubble-tension witness in a fixed same-target cosmological frame has nonempty image in the AASC cosmological normal-form family $N_1, \dots, N_7$.*

Proof. By the cosmological divergence theorem, at least one technical coordinate L_i is nontrivial. The displayed table defines a total mapping from $L_1, \dots, L_7$ to $N_1, \dots, N_7$. Therefore the normal-form image is nonempty. $\square$

Theorem 6.3 (Collapse of skin and same-scope repair). *No member of the normal-form family can be treated as a harmless same-scope redescription. Each nontrivial normal form is either an explicit anchor/tensor/quotient/continuation/boundary/error change or an estimator selector. Therefore a persistent Hubble-tension witness marks certificate incompleteness or theorem-object change, not a purely presentational discrepancy.*

Proof. Anchor motion changes the comparison domain. Tensor-witness change changes standing-bearing content. Quotient motion changes the target. Continuation failure changes lawful early-to-late transport. Boundary-record failure changes the standing of measurement records. An estimator selector performs gate or classifier work not fixed by the quotient architecture. Error-envelope motion changes the comparison rule itself. None is skin. If a purported divergence changes none of these, it is type-preserving and cannot produce persistent noncommutation by the divergence theorem. $\square$

7 Hubble tension theorem

Theorem 7.1 (Hubble tension as transport divergence). *Let $\mathcal{I}_{\Lambda\text{CDM}}$ be treated as a fixed-domain cosmological bookkeeping realization with early and late expansion-inference protocols. If a persistent early/late Hubble-tension witness exists and the two estimates are claimed to estimate the same quotient expansion target, then ΛCDM is not a complete constraint-native realization certificate for that comparison. At least one component of the cosmological construction type is incomplete, moved, or implicitly supplemented.*

Proof. A complete constraint-native realization certificate contains a same-domain well-typed early-to-late transport for the relevant comparison. If ΛCDM supplied such a certificate and the two estimates referred to the same quotient target, the cosmological divergence theorem would force commutation of early and late inference within the declared error envelope. A persistent Hubble-tension witness violates that commutation. Therefore the certificate is not complete for that comparison. The normal-form image theorem identifies the possible failure loci. $\square$

Proposition 7.2 (ΛCDM transport-incompleteness diagnostic). *In the fixed-domain reading used here, base- ΛCDM becomes incomplete for a persistent Hubble-tension witness exactly when it treats the CMB-derived estimate and the late-distance estimate as same-target estimates while failing to supply a single admissibility-preserving transport through the seven certificate components. The incompleteness is not the mere fact that ΛCDM has parameters; it is the failure of the claimed common H_0 target to commute through the declared early and late inference pipelines.*

Proof. The external ΛCDM inference object already contains the model family, two pipelines, record structures, and error disciplines. If these data supply a same-domain transport, the divergence theorem gives commutation. If persistent noncommutation remains while the same target is asserted, then the missing item is not a new verbal interpretation of ΛCDM but a missing or moved certificate component: anchor, tensor witness, quotient target, continuation, boundary record, estimator, or error envelope. That is precisely transport incompleteness. $\square$

Remark 7.3 (What the theorem does not say). *The theorem does not say which ledger is incomplete and does not derive the observed discrepancy. It says that a same-target persistent discrepancy cannot be merely notational. It must be anchor-level, tensor-level, quotient-level, continuation-level, boundary/record-level, estimator-level, or error-envelope-level. Dynamical model-building then becomes the task of supplying a revised certificate.*

8 Concrete cosmological witnesses

8.1 Hidden tensor witness toy model

Definition 8.1 (Two-epoch toy ledger). *Let an early witness be $w_E = (r_s, \Omega_m, \Omega_\Lambda)$, where r_s is a sound-horizon witness. Let a late witness be $w_L = (d_L(z), \mathcal{C})$, where d_L is a luminosity-distance witness and $\mathcal{C}$ is a calibration record. Let*

$$I_E(w_E) = H_*, \quad I_L(w_L) = \frac{z}{d_L(z)}$$

for small z , suppressing higher-order terms and units.

Proposition 8.2 (Explicit mismatch from an omitted tensor witness). *Suppose the actual late witness is*

$$d_L(z) = \frac{z}{H_*} (1 + \gamma\chi)$$

where χ is a tensor witness absent from the declared ledger and $\gamma \neq 0$. Then the declared early and late estimators give

$$I_E(w_E) = H_*, \quad I_L(w_L) = \frac{H_*}{1 + \gamma\chi}.$$

If $\gamma\chi \neq 0$ beyond the declared error envelope, the mismatch is persistent and has normal-form image N_2 or N_4 , not skin.

Proof. The computation is immediate from the displayed estimators. The mismatch is not a coordinate artifact because all declared skin variables have been held fixed. The missing object χ either belongs in the tensor ledger as an expansion-relevant witness, giving N_2 , or changes late continuation without being represented, giving N_4 . In neither case is it a harmless redescription. $\square$

Proposition 8.3 (Completed toy certificate restores commutation). *In the same toy ledger, extend the tensor ledger by adding χ , and replace the late estimator by the certificate-corrected estimator*

$$I_L^{\text{corr}}(w_L, \chi) := (1 + \gamma\chi) I_L(w_L).$$

Then

$$I_L^{\text{corr}}(w_L, \chi) = H_* = I_E(w_E).$$

Thus the mismatch is removed by completing the tensor ledger and modifying the estimator certificate, not by reparametrizing the old ledger.

Proof. Substituting $I_L(w_L) = H_*/(1 + \gamma\chi)$ gives

$$I_L^{\text{corr}}(w_L, \chi) = (1 + \gamma\chi) \frac{H_*}{1 + \gamma\chi} = H_*.$$

The corrected comparison is not skin relative to the original certificate because χ is a newly declared tensor witness and the estimator ledger has been revised. It is a certified completed comparison, exactly as the realization-certificate theorem requires. $\square$

Remark 8.4 (Interpretation of the toy model). *The symbol χ is not a proposed physical field. It represents any omitted standing-bearing tensor witness: late-time equation-of-state behavior, calibration-relevant boundary data, vacuum-sector contribution, perturbation witness, or record-lineage correction. The example shows the algebraic form of transport failure and certificate completion, not a preferred cosmological mechanism.*

8.2 Estimator-selector witness

Proposition 8.5 (Prior/estimator mismatch as selector). *Let the same likelihood family $\mathcal{L}(d \mid \theta)$ be evaluated with two priors π_E and π_L , producing expansion estimates*

$$\hat{H}_E = \text{Est}(\mathcal{L}_E, \pi_E), \quad \hat{H}_L = \text{Est}(\mathcal{L}_L, \pi_L).$$

If the priors are not fixed by the estimator ledger and their difference changes the reported same-target H_0 , then the discrepancy has normal-form image N_6 .

Proof. A prior or estimator choice that changes the reported expansion target while not being fixed by the quotient ledger performs standing-bearing work. It therefore acts as an estimator selector. If the prior choice is declared part of the estimator ledger, the certificate has changed; if it is not declared, it is unlicensed same-scope selector work. $\square$

8.3 Boundary-record witness

Proposition 8.6 (Calibration-record mismatch). *Let $\mathcal{C}_E$ and $\mathcal{C}_L$ be calibration records used in early and late inference. If $\mathcal{C}_E$ and $\mathcal{C}_L$ are claimed to represent the same calibration anchor but no record transport $T_R(\mathcal{C}_E) = \mathcal{C}_L$ is supplied, then a resulting expansion discrepancy has normal-form image N_5 or N_1 .*

Proof. If the calibration records are same-anchor records, a record transport must preserve their standing role. Without such transport, the mismatch is boundary-record failure. If the records are not same-anchor records, the anchor ledger has moved. These are N_5 and N_1 , respectively. $\square$

9 Witness ledger for common resolution classes

The following table is diagnostic rather than exhaustive. It shows how familiar Hubble-tension resolution strategies are typed by the AASC ledger.

Resolution class	AASC ledger role	Required certificate item
Early dark energy / early expansion modification	Tensor-witness or continuation change	New early-time energy witness and proof of transport into late expansion target
Time-varying dark energy $w(z)$	Tensor-witness and continuation change	Equation-of-state witness with lawful continuation and matching to CMB, BAO, SNe
Extra relativistic species / dark radiation	Tensor-witness change	Additional radiation witness with early/late transport and perturbation compatibility
Modified gravity	Anchor or continuation change	New GR-projection theorem object or modified continuation ledger; cannot be skin
Local void / environmental effect	Anchor or quotient-target risk	Proof that local inferred H_0 and global H_0 are same target, or explicit quotient motion
Calibration/systematic correction	Boundary-record or error-envelope change	Updated record ledger, calibration anchors, and error discipline
Estimator/prior revision	Estimator selector risk	Proof that estimator change is skin, or declaration of altered inference certificate

Proposition 9.1 (Diagnostic non-neutrality). *Every resolution class in the table changes at least one component of the realization certificate unless it is pure skin. If it is pure skin, it cannot resolve a persistent same-target discrepancy.*

Proof. Each row names the certificate component altered by the proposed resolution. If the proposed modification alters no certificate component, then it preserves the construction type and cannot alter the same-target inference by the divergence theorem. If it does alter one, it is a typed resolution candidate rather than a harmless redescription. $\square$

10 Realization-certificate theorem

Definition 10.1 (AASC cosmological resolution). *An AASC cosmological resolution of a persistent Hubble-tension witness is a revised realization certificate*

$$\text{Cert}'_C = (\tau'_C, T'_C)$$

such that early and late inference commute for the witness pair, and the revision explicitly identifies which normal-form locus was changed.

Theorem 10.2 (Realization-certificate theorem). *A proposed resolution is admissible as a same-target AASC cosmological resolution if and only if it supplies:*

1. *a fixed anchor ledger identifying the comparison scope;*
2. *a tensor-witness ledger sufficient for early and late expansion observables;*
3. *a quotient-identity ledger specifying what counts as the same expansion target;*
4. *a continuation ledger from early to late epochs;*
5. *a boundary/record ledger for observational and calibration records;*
6. *an estimator ledger for inference maps and priors;*
7. *an error-envelope ledger for statistical and systematic uncertainty;*
8. *a commutation proof showing early and late inference agree within the revised error discipline.*

Without such a certificate, the proposal is not a complete AASC resolution; it is a dynamical suggestion, calibration suggestion, or scope change awaiting admissibility classification.

Proof. Necessity follows from the divergence theorem: each ledger component is a possible first divergence locus, so omitting any component leaves a possible untyped failure. Sufficiency follows because a complete ledger with a commutation proof supplies admissibility-preserving transport of the same expansion target across early and late inferences. The revised certificate may change scope or tensor content, but it does so explicitly rather than as same-scope repair. $\square$

11 Existing cosmological programs as diagnostic sketches

This section is not a technical review of every cosmological model. It records how common programmatic strategies are typed by the certificate theorem.

Early-universe solutions. Early dark energy, altered sound-horizon physics, or additional relativistic species typically act on L_2 and L_4 : tensor witnesses and continuation. To count as an AASC resolution, such a model must show not only that it shifts the CMB-inferred H_0 , but that the new tensor witness transports consistently into the late-universe quotient target.

Late-universe solutions. Dynamical dark energy, interacting dark sectors, environmental effects, or redshift-dependent expansion histories act primarily on L_2 , L_3 , and L_4 . They must show whether the late-time inferred quantity is the same quotient target as the early-time inferred H_0 , or else declare quotient motion.

Modified-gravity solutions. Modified gravity is not a skin change. It modifies the GR projection, continuation ledger, or anchors. Therefore it must supply a new constrained-Hamiltonian or effective-field realization certificate, not merely a modified Friedmann equation.

Calibration and systematics. Calibration revisions, Cepheid/TRGB standardization, supernova selection, and record-level systematics act on L_5 and L_7 . If they resolve the mismatch, they do so by changing the boundary-record or error-envelope ledger, not by changing cosmological tensor content.

Estimator and prior changes. Changes in priors, likelihoods, or parametrizations act on L_6 unless proven to be skin. Such changes are admissible only if they either preserve the quotient target and error discipline or explicitly revise the estimator certificate.

12 Relation to data and empirical work

This paper does not replace empirical cosmology. It gives a structural audit. Planck, SH0ES, BAO, DESI, JWST cross-checks, TRGB, strong-lensing, and standard-siren programs supply the witness data. AASC supplies a classification of what a successful same-target comparison must preserve. The theorem therefore supports, rather than displaces, empirical work: it says exactly what a proposed resolution must certify.

Remark 12.1 (Why no number is derived). *The observed value of the Hubble tension depends on data, calibrations, likelihoods, and cosmological modeling. AASC does not predict a number without a realization certificate. It proves that a persistent number mismatch cannot be a mere same-scope skin effect.*

13 Conclusion

The Hubble tension can be treated as a theorem-level transport problem rather than as a purely dynamical puzzle. If early and late expansion inferences are same-target claims inside a complete AASC cosmological realization, their transport ledger must commute. If they do not commute, the divergence has a nonempty image in a finite family of typed loci: anchors, tensor witnesses, quotient identity, continuation, boundary records, estimators, or error envelopes.

This does not solve the tension. It makes the burden exact. A successful solution must supply a realization certificate connecting early and late epochs as projections of the same admissibility-bearing quotient architecture. In that sense, the framework replaces the question “which patch fixes H_0 ?” with the sharper question: which revised cosmological certificate makes the early and late expansion targets the same standing-bearing object under admissible transport?

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